Assignment 4: Implementing Control Structures

Course: MSCS-632 Advanced Programming Languages

Assignment: Employee Shift Scheduling System

Programming Languages: Go and Python

GitHub Repository

https://github.com/tiwariarjun91/MSCS-632-Assignment-4-Implementing-Control-Structures

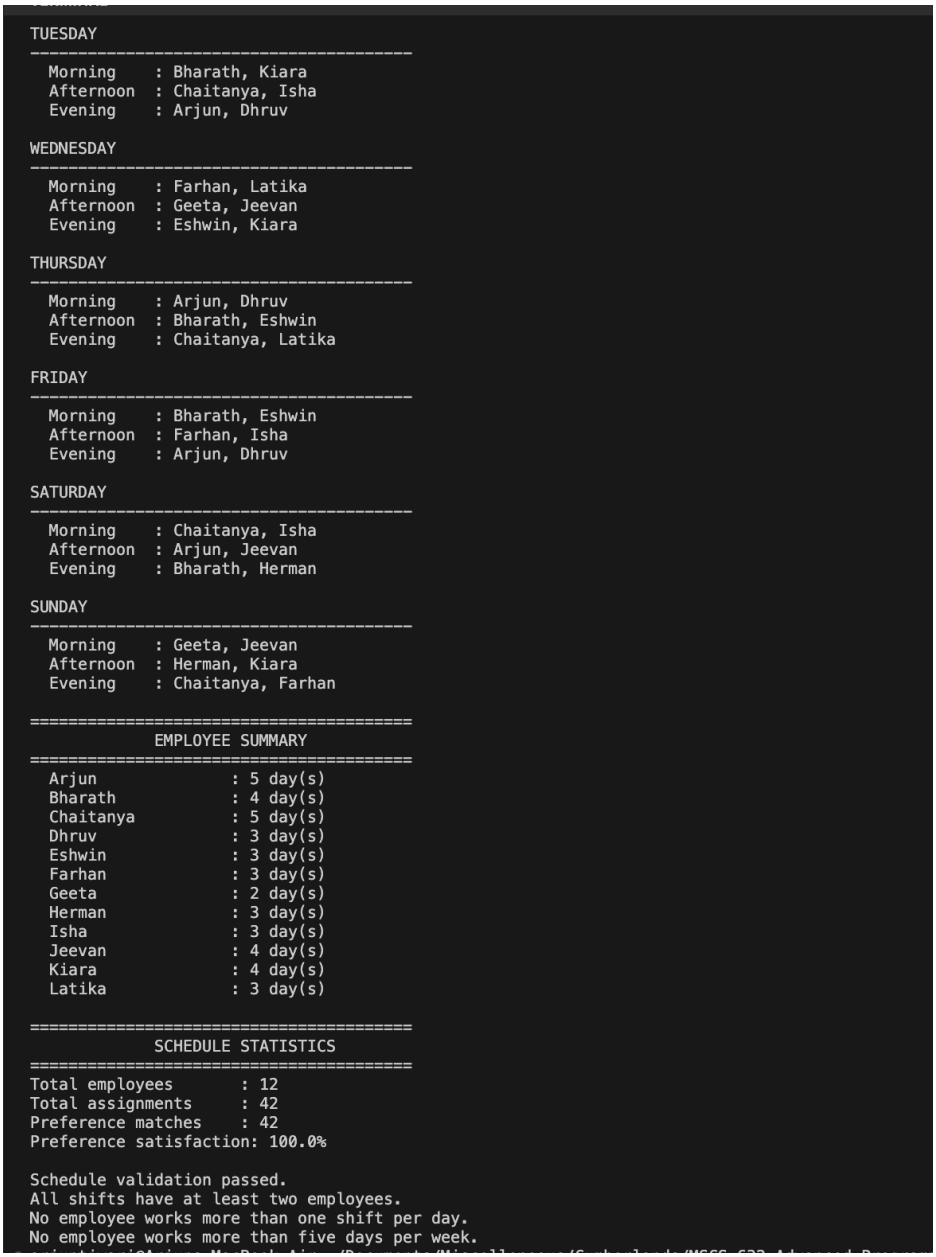
The repository contains both Go and Python implementations, source code, sample employee data, execution instructions, and screenshots.

Go Implementation — Output Screenshots

Screenshot 1:



Screenshot 2:



```
TUESDAY
----------------------------------------
  Morning    : Bharath, Kiara
  Afternoon  : Chaitanya, Isha
  Evening    : Arjun, Dhruv

WEDNESDAY
----------------------------------------
  Morning    : Farhan, Latika
  Afternoon  : Geeta, Jeevan
  Evening    : Eshwin, Kiara

THURSDAY
----------------------------------------
  Morning    : Arjun, Dhruv
  Afternoon  : Bharath, Eshwin
  Evening    : Chaitanya, Latika

FRIDAY
----------------------------------------
  Morning    : Bharath, Eshwin
  Afternoon  : Farhan, Isha
  Evening    : Arjun, Dhruv

SATURDAY
----------------------------------------
  Morning    : Chaitanya, Isha
  Afternoon  : Arjun, Jeevan
  Evening    : Bharath, Herman

SUNDAY
----------------------------------------
  Morning    : Geeta, Jeevan
  Afternoon  : Herman, Kiara
  Evening    : Chaitanya, Farhan

========================================
            EMPLOYEE SUMMARY
========================================
  Arjun              : 5 day(s)
  Bharath            : 4 day(s)
  Chaitanya          : 5 day(s)
  Dhruv              : 3 day(s)
  Eshwin             : 3 day(s)
  Farhan             : 3 day(s)
  Geeta              : 2 day(s)
  Herman             : 3 day(s)
  Isha               : 3 day(s)
  Jeevan             : 4 day(s)
  Kiara              : 4 day(s)
  Latika             : 3 day(s)

========================================
            SCHEDULE STATISTICS
========================================
Total employees       : 12
Total assignments     : 42
Preference matches    : 42
Preference satisfaction: 100.0%

Schedule validation passed.
All shifts have at least two employees.
No employee works more than one shift per day.
No employee works more than five days per week.
```

Screenshot 3:

```
  Evening    : Arjun, Dhruv

WEDNESDAY
----------------------------------------
  Morning    : Farhan, Latika
  Afternoon  : Geeta, Jeevan
  Evening    : Eshwin, Kiara

THURSDAY
----------------------------------------
  Morning    : Arjun, Dhruv
  Afternoon  : Bharath, Eshwin
  Evening    : Chaitanya, Latika

FRIDAY
----------------------------------------
  Morning    : Bharath, Eshwin
  Afternoon  : Farhan, Isha
  Evening    : Arjun, Dhruv

SATURDAY
----------------------------------------
  Morning    : Chaitanya, Isha
  Afternoon  : Arjun, Jeevan
  Evening    : Bharath, Herman

SUNDAY
----------------------------------------
  Morning    : Geeta, Jeevan
  Afternoon  : Herman, Kiara
  Evening    : Chaitanya, Farhan

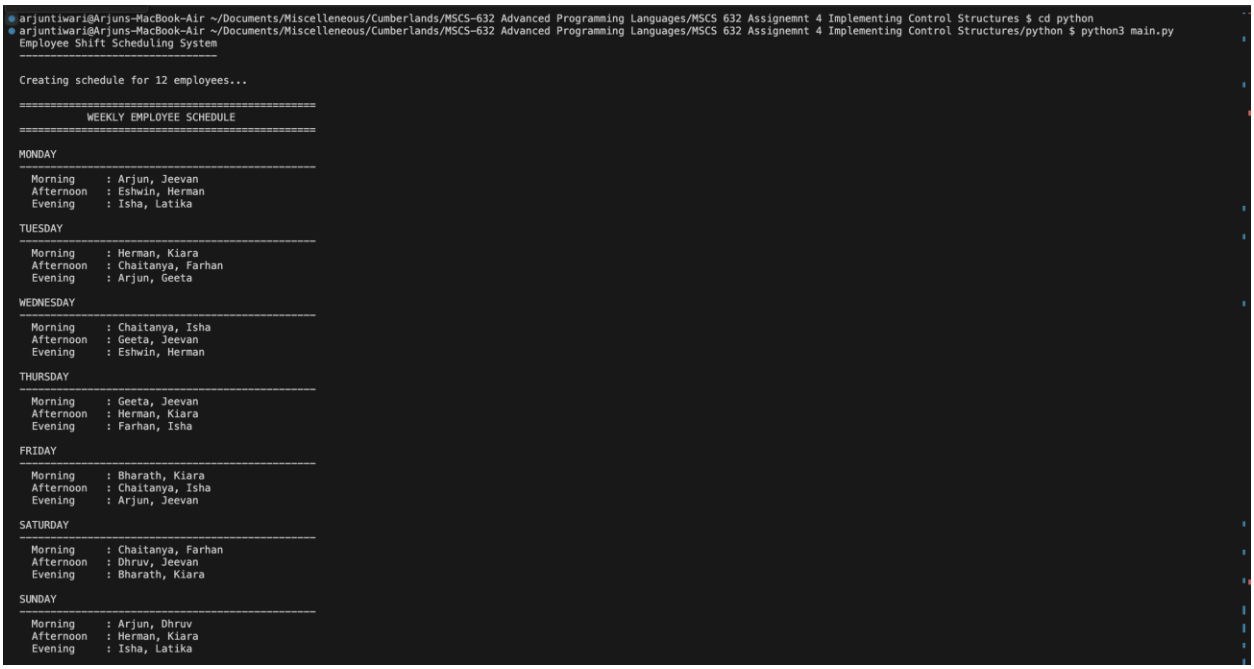
========================================
            EMPLOYEE SUMMARY
========================================
  Arjun              : 5 day(s)
  Bharath            : 4 day(s)
  Chaitanya          : 5 day(s)
  Dhruv              : 3 day(s)
  Eshwin             : 3 day(s)
  Farhan             : 3 day(s)
  Geeta              : 2 day(s)
  Herman             : 3 day(s)
  Isha               : 3 day(s)
  Jeevan             : 4 day(s)
  Kiara              : 4 day(s)
  Latika             : 3 day(s)

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```

Python Implementation — Output Screenshots

Screenshot 1:



Screenshot 2:



Conclusion

Both implementations successfully generate a weekly employee shift schedule while enforcing staffing requirements, employee work limits, and shift preferences.